

Exp No: 2

GENERATION OF SQUARE WAVE USING PROTEUS

AIM:

To write an assembly language program to Generate square wave using AT89C51.

SOFTWARE REQUIRED:

- Proteus 8 software.

PROGRAM

```
ORG 0000H
```

```
UP: SETB P2.0
```

```
ACALL DELAY
```

```
CLR P2.0
```

```
ACALL DELAY
```

```
SJMP UP
```

```
DELAY: MOV R4,#35
```

```
H1:MOV R3,#255
```

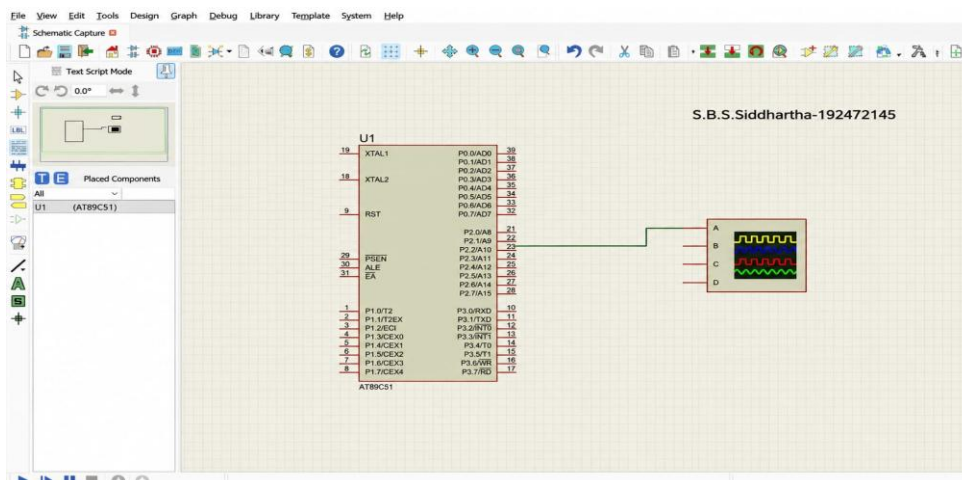
```
H2:DJNZ R3,H2
```

```
DJNZ R4,H1
```

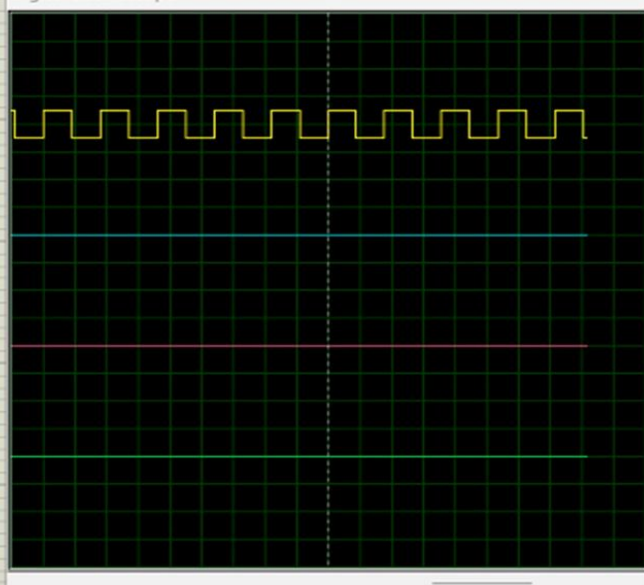
```
RET
```

```
END
```

Output:



Digital Oscilloscope



Trigger

Level: -10, 0, 10

AC DC

Auto

One-Shot

Cursors

Source: A, B, C, D

Channel A

Position: 110, 120, 130

AC DC GND OFF

Invert

A+B

0.5 0.2 0.1

2 5 10 20 50 100

V 5 mV

Channel C

Position: 50, 40, 30

AC DC GND OFF

Invert

C+D

0.5 0.2 0.1

2 5 10 20 50 100

V 5 mV

Horizontal

Source: A, B, C, D

Position: 210, 200, 190

0.5 0.2 0.1

2 5 10 20 50 100

ms 20m μ s

Channel B

Position: 30, 40, 50

AC DC GND OFF

Invert

0.5 0.2 0.1

2 5 10 20 50 100

V 5 mV

Channel D

Position: 130, 120, 110

AC DC GND OFF

Invert

0.5 0.2 0.1

2 5 10 20 50 100

V 5 mV